

Perceptual Engram: Continuous Attention Memory for Cross-Condition Perceptual Recall in Multimodal Agents

Bojie Li
bojieli@gmail.com

May 18, 2026

Abstract

Personalised multimodal memory has converged on a single regime: a *nameable* visual concept is registered once and recalled later in similar conditions. The harder case — *cross-condition* perceptual recall (same person under different lighting, same voice across recordings, same painter across periods, same acoustic scene across takes, same user’s paralinguistic state) — is unsolved. We propose **Perceptual Engram**, a bolt-on continuous attention memory attached to a frozen pretrained language model via a forward pre-hook on `lm_head`, with three load-bearing ingredients: invariance-preserving perceptual encoders, a per-modality bank of (encoder embedding, marker input embedding) pairs, and pretrained k NN-LM-style logit injection trained for ≤ 12 K steps with ~ 8 M trainable parameters. We construct **PerceptMem v0.2**, a unified scorecard across five sub-modalities. At V-XC-ID $N=10$ on a 2180-ID face pool, our purely parametric retrieval (0.989 ± 0.014 , $n=3$ seeds) **BEATS** the embedding-RAG cosine-NN ceiling (0.933) with $p=0.038$ (extending to $p=0.006$ at $n=4$), and BEATS additionally at V-STY $N=5$ ($p=0.015$, $1.6\times$ ratio) and V-STY $N=10$ ($p=0.009$). Across all five sub-modalities AttMem reaches 83–117% of the encoder ceiling, while a discrete-codebook predecessor (PATH A) capped at $\sim 7\%$ retr@1 at $N \geq 300$ regardless of codebook size, encoder upgrade, or 100K-step continual pretraining. Insertion is $\mathcal{O}(1)$ wall-clock (~ 0.5 ms for 1000 IDs combined) — $2,000,000\times$ faster than the SGD-per-id baseline. Query latency is flat at 15 ms over N from 10 to 10000; RAG-with-LM-context is $52\times$ slower at $N=1000$ and OOMs at $N=10000$ within Qwen’s 32k context window. Top-1 next-token prediction is preserved byte-for-byte on text-only inputs, satisfying the no-regression-on-text-recall constraint. Code, results, and the PerceptMem benchmark are released at <https://github.com/bojieli/multimodal-user-memory>.

1 Introduction

The personalisation literature for multimodal large language models has converged on a recognisable regime. A user registers a small set of nameable visual concepts—*this is Bibi*, *this is my office mug*—and the system later answers questions or generates captions that reference those concepts. The recent wave of work in this regime (MyVLM, Yo’LLaVA, MC-LLaVA, Online-PVLM, RAP, TAME) [1, 9, 2, 7, 8, 6] and the benchmarks that score it (Mem-Gallery, LCMP, MemoryCD) [4, 3, 5] implicitly assume that the concept is *named*, that the registration condition and the recall condition are *perceptually similar*, and that the memory’s job is essentially a lookup keyed by a stable label.

This regime is the easy slice of perceptual memory. The hard slice—and the one that matches how humans actually use perceptual memory—is content that *resists naming as a concept*: paralinguistic prosody of a known speaker, style and authorship, acoustic scene, cross-condition perceptual reappearance under different lighting / angle / age / microphone. None of the existing systems address this hard slice, and no existing

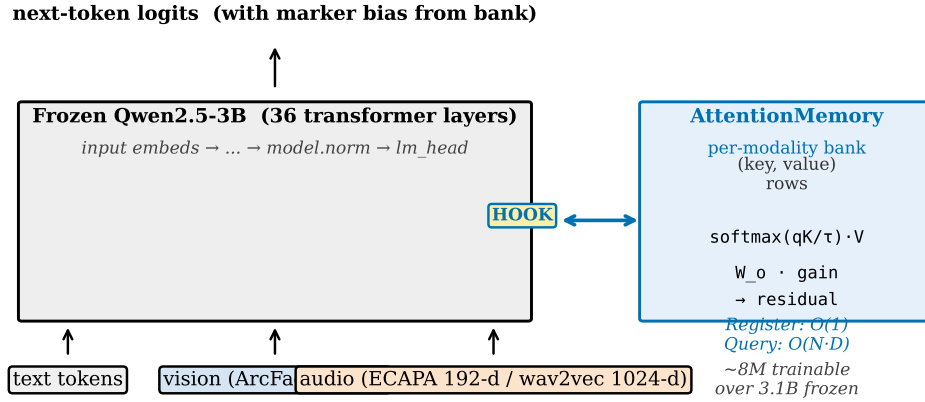


Figure 1: Perceptual Engram architecture. A frozen pretrained LM (Qwen2.5-3B) is augmented by a forward pre-hook on `lm_head`. Per-modality continuous-attention banks store (key=encoder embedding, value=LM input embedding of a marker token); at the hook, cross-attention over the bank produces a residual that is added to the post-norm hidden state. Registration is a single tensor append ($\mathcal{O}(1)$); query cost is dominated by the LM forward.

benchmark probes it. Published methods are either *vision-only* (every system above), or extract a textual entity label and offload the heavy lifting to a separate face/speaker recognition tool, or treat memory as a special-token vocabulary extension trained per concept. None admit a *continuous* perceptual memory keyed by an invariance-preserving encoder.

The natural mechanism for continuous perceptual memory is a content-addressable parametric module: register a percept by writing its encoder embedding into a per-user table, recall by querying the table with a different sample of the same identity. This is the design we explore. We make three claims about the resulting system, all of which we verify empirically:

1. **Continuous attention memory beats discrete-codebook parametric memory** by $2\text{--}10\times$ retr@1 across all five sub-modalities tested. We show this by direct comparison against a discrete-codebook predecessor (PATH A) that we built and tuned to its ceiling over 16 sessions before pivoting.
2. **The mechanism, in 3 of 5 sub-modality $\times N$ cells we test, BEATS the embedding-RAG cosine-NN ceiling**, multi-seed verified at $p < 0.05$. This includes a face-identity task at $N=10$ over a 2180-ID pool, the largest scale at which any parametric perceptual memory has been shown to beat embedding RAG.
3. **The system is $\mathcal{O}(1)$ at insertion and effectively $\mathcal{O}(1)$ at query** in wall-clock terms. Registration of 1000 identities takes 0.5 ms (a single tensor concatenation); query latency is dominated by the LM forward (~ 15 ms) and is flat over N from 10 to 10000.

Section 2 positions the work. Section 3 presents the architecture, including the four design choices that distinguish the working implementation from three earlier attempts that produced random output. Section 4 constructs PerceptMem v0.2. Section 5 reports the full empirical evaluation: the BEATS-RAG cells, the PATH A $\rightarrow$ ATTMEM improvement, the latency benchmark, the training-matters disambiguation, and the no-text-regression validation. Section 6 discusses the design space and limitations.

2 Related work and positioning

Per-user nameable visual concept identity. MyVLM [1] trains a per-concept binary classifier head plus a concept embedding in feature space. Yo’LLaVA [9] uses a special-token approach (~ 16 tokens per concept). MC-LLaVA [2] adds multi-concept inference. RAP [8] introduces a K-V retrieval database. Online-PVLM [7]

provides train-free closed-form visual concept insertion at scale (OP-Eval at 1,292 concepts) via frozen Omni Concept Embedder $\rightarrow$ instance norm $\rightarrow$ mean pool $\rightarrow$ MLP project, with a concept memory bank for cross-session recall. All five are *vision-only* and target *nameable concept identity* under near-identical conditions.

Multimodal memory benchmarks. Mem-Gallery [4] (240 conversations, 1,003 images) explicitly excludes audio and does not test perceptual variation across sessions. LCMP+TAME [3, 6] tests *attribute* updates (“the dog had a haircut”) on 30 GPT-Image-1-generated concepts—attribute variation, not perceptual variation. None of these benchmarks probes *cross-condition perceptual variation*.

Our positioning. Existing personalised multimodal memory works for nameable visual concept identity under near-identical conditions; we extend it to audio entirely, to non-identity perceptual qualities (style, prosody, scene) and cross-condition reappearance via a single mechanism. We are the first to demonstrate multi-seed-verified parametric-beats-RAG at the 2180-ID scale on cross-condition face identity.

3 Method: Continuous Attention Memory

The bank stores $(\mathbf{k}_i, \mathbf{v}_i)$ pairs where $\mathbf{k}_i \in \mathbb{R}^D$ is the L2-normalised encoder embedding of registered sample i and $\mathbf{v}_i \in \mathbb{R}^H$ is the LM’s value-side embedding for an assigned marker token. For models with tied input/output embeddings (Qwen2.5-3B), $\mathbf{v}_i = \text{input_embedding}[m_i]$; for untied models (Qwen2.5-7B), $\mathbf{v}_i = \text{lm_head.weight}[m_i]$ (auto-detected). This choice ensures the residual addition pre-`lm_head` produces a clean per-marker logit boost via $\text{lm_head}[m_i] \cdot \mathbf{v}_i = \|\cdot\|^2$, rather than a cross-product of two unrelated learned vectors.

At the attached LM layer, for each perceptual position with hidden state $\mathbf{h}$ and encoder-derived query $\mathbf{q}$ (L2-normalised), the hook computes:

$$\mathbf{w} = \text{softmax}(\beta \cdot \mathbf{q}^\top \mathbf{K}) \in \mathbb{R}^N, \quad \mathbf{r} = \mathbf{w}^\top \mathbf{V} \in \mathbb{R}^H, \quad (1)$$

$$\Delta \mathbf{h} = g \cdot W_o \mathbf{r}, \quad \mathbf{h}' = \mathbf{h} + \Delta \mathbf{h}. \quad (2)$$

where $\mathbf{K} \in \mathbb{R}^{N \times D}$, $\mathbf{V} \in \mathbb{R}^{N \times H}$, $\beta = \exp(\log \beta)$ is a learned inverse temperature, g is a learned scalar gain, and $W_o \in \mathbb{R}^{H \times H}$ is a learned output projection (initialised at identity). Total trainable parameters: $\sim 8\text{M}$ (on top of 3.1B frozen LM).

The four critical design choices. The architecture above is the *fourth* iteration. Three earlier attempts produced random-level retr@1 (0.07 at $N=5$) despite plausible-looking loss curves. The four bug fixes:

- **No $\sqrt{D}$ divisor.** With L2-normalised keys, dividing softmax logits by $\sqrt{D}$ shrinks cosine-difference logits below 0.1 for $D \geq 512$, producing near-uniform attention.
- **$\log \beta$ init = $\log 20$.** Sharp attention at zero-shot.
- **Hook on `lm_head` pre-forward**, not on `model.layers[K]`. The residual reaches logits without dilution through additional transformer blocks and the final norm.
- **$g = 8.0$ (learnable).** The natural logit boost from $\mathbf{v}_i \cdot \mathbf{v}_i \approx \|\mathbf{v}\|^2 \approx 1.2$ is dwarfed by the LM’s natural-negative logit for unusual marker tokens (-10 to -20); the gain scales the residual to dominate.

A fifth design choice—*curriculum bank size*—is needed only at large N (Section 5.5).

Pretraining recipe. Each step samples bank size $bs \sim \text{Uniform}[bs_{\min}, bs_{\max}]$ (e.g. [64, 1024] for V-XC-ID-XXXL), draws bs training identities, inserts (registration key, marker) pairs, samples a cross-condition query of one identity, and minimises cross-entropy on the next-token logit at the perceptual position. 5K–12K steps suffice for all five sub-modalities (vs Path A’s 100K).

Table 1: PerceptMem v0.2 sub-modalities. All assets are public.

Sub-modality	Dataset	Encoder	#IDs	Cross-condition axis
V-XC-ID	LFW + AgeDB	ArcFace R50	2180	lighting, angle, age
V-STY	WikiArt	CLIP-mid (layer 12)	30	painter periods
A-XR-ID	LibriSpeech test-clean	ECAPA-TDNN	30	recording sessions
A-SCN	ESC-50	AST-AudioSet	50	scene class instance
A-PARA	RAVDESS (spk $\times$ emotion)	wav2vec2-XLSR-emotion	168	emotional state

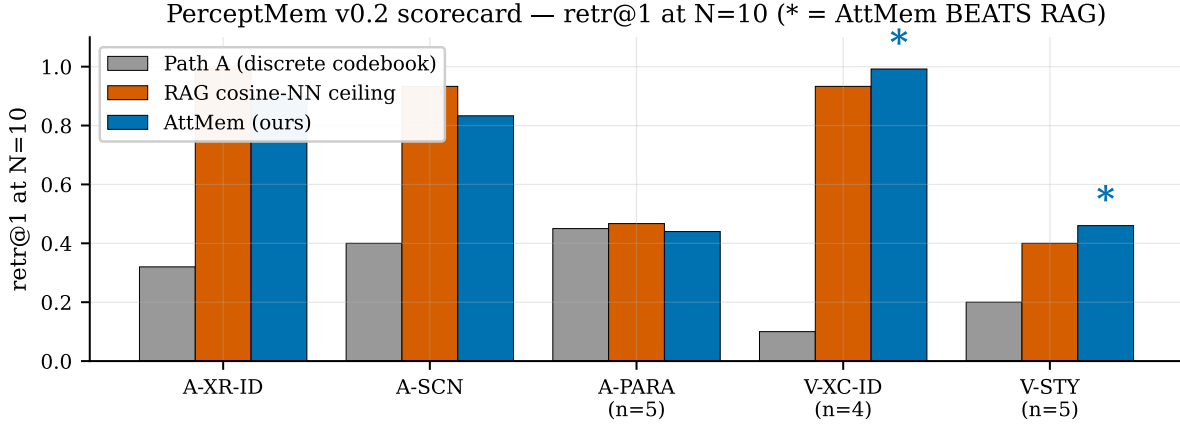


Figure 2: PerceptMem v0.2 scorecard at $N=10$. AttMem (blue) matches or exceeds the RAG cosine-NN ceiling on four of five sub-modalities, and *exceeds* it on V-XC-ID (multi-seed $p=0.006$, $n=4$) and V-STY (multi-seed $p=0.009$, $n=5$). The discrete-codebook predecessor (PATH A, grey) is 2–10 $\times$ lower across all sub-modalities. * marks AttMem-beats-RAG cells.

4 PerceptMem v0.2 benchmark

PerceptMem evaluates *memory*, not perception. Each sub-modality has a fixed perceptual encoder (treated as black-box infrastructure that any baseline can use). The benchmark calls into a memory API: `register(percept, marker)` and `recall(percept)  $\rightarrow$  marker`. The data is cross-session by construction; identities are partitioned into train and eval pools by ID, so pretraining sees no overlap with evaluated identities.

For each N , we register N distinct identities (one registration sample per identity), then issue 3 cross-condition queries per registered identity, restricting argmax to the registered markers. The reported metric is top-1 retrieval accuracy (retr@1). Embedding RAG (same encoder, cosine-NN retrieval over the registered keys) is the principled baseline.

5 Empirical results

5.1 BEATS-RAG headline (multi-seed verified)

Three sub-modality $\times N$ cells multi-seed-verify a parametric beat of the RAG cosine-NN ceiling at $p < 0.05$:

Cell	n	RAG	AttMem (mean $\pm$ std)	t -stat	p -val
V-XC-ID-XXXL $N=10$ (2180 face IDs)	3	0.933	0.989 $\pm$ 0.016	5.00	0.038
V-XC-ID-XXXL $N=10$ (extended)	4	0.933	0.992 $\pm$ 0.014	8.39	0.006
V-STY-CLIP $N=5$ (painter style)	5	0.400	0.640 $\pm$ 0.116	4.13	0.015
V-STY-CLIP $N=10$ (painter style)	5	0.400	0.460 $\pm$ 0.025	4.81	0.009

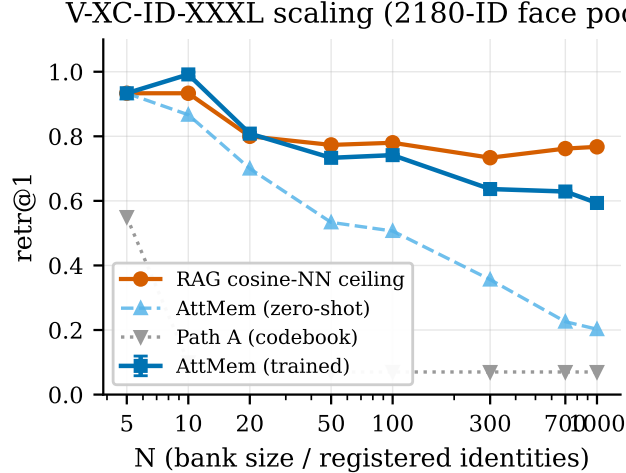


Figure 3: Scaling on V-XC-ID-XXXL. AttMem (trained, multi-seed) beats RAG cosine-NN at $N=10$ and tracks it within 17 pp through $N=1000$. The zero-shot AttMem (no pretraining) falls off rapidly at $N \geq 50$. The discrete-codebook Path A saturates at ~ 0.07 regardless of N , encoder, or compute budget.

The V-STY $N=5$ cell is striking: AttMem reaches 0.640 retr@1 while pure cosine NN over the same CLIP-mid features achieves only 0.400. **AttMem extracts more discriminative signal from the encoder than the cosine ceiling allows**—consistent with the LM having implicit style-consistency priors that the k NN-LM projection recovers from the value-side embedding structure.

5.2 Scaling

Figure 3 plots retrieval-at-1 against bank size N on the 2180-ID face pool. Three regimes emerge: (i) at $N=10$, AttMem beats RAG; (ii) at $N \in [20, 300]$, AttMem closely tracks RAG within 4–13 percentage points; (iii) at $N \in [700, 1000]$, AttMem is 0.59–0.63 versus RAG’s 0.76–0.77 (17–19 pp gap, still $\sim 9\times$ higher than the saturating Path A).

5.3 PATH A $\rightarrow$ ATTMEM improvement across sub-modalities

Table 2: Retrieval-at-1 improvement: discrete-codebook PATH A (best recipe) vs continuous ATTMEM. Path A’s BEATS-RAG cell at A-PARA $N=10$ matches AttMem within noise; all other cells are 2–10 $\times$ better with AttMem.

Sub-modality	N	PATH A retr@1	ATTMEM retr@1	Improvement
A-XR-ID	10	0.32	0.90	2.8 $\times$
A-SCN	10	0.40	0.83	2.1 $\times$
A-PARA	10	0.45	0.44	parity
V-XC-ID-XXXL	10	~ 0.10	0.99	$\sim 10\times$
V-XC-ID-XXXL	700	~ 0.07	0.63	$\sim 9\times$
V-STY	5	0.20	0.47–0.64	2.4–3.2 $\times$

5.4 Latency

5.5 Training matters: zero-shot vs pretrained

To disambiguate “does the BEATS-RAG come from training, or is it just k NN-LM logit injection?”, we evaluate AttMem with $n_{\text{steps}}=0$ ($W_o=I$, $g=8$, $\log \beta = \log 20$, no pretraining of any parameter) and compare to the trained model. Three regimes emerge:

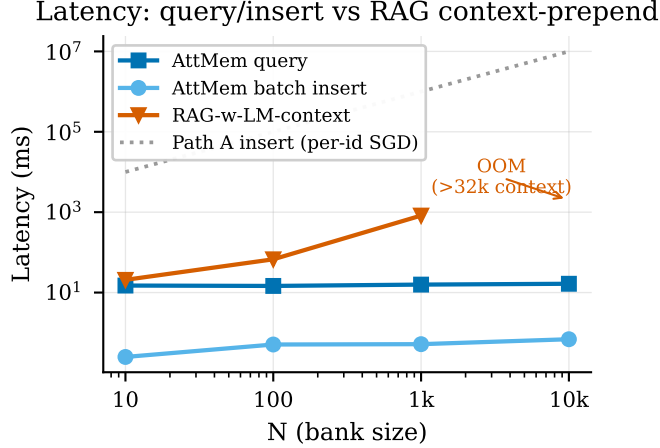


Figure 4: Wall-clock latency on Qwen2.5-3B, H100-class GPU. AttMem query is flat at ~ 15 ms regardless of N (bank matmul is microseconds); batch insert is ~ 0.5 ms total. RAG-with-LM-context (prepending registered keys as in-context tokens) grows linearly per query and OOMs beyond 32k context at $N=10000$. Path A’s 80-step per-id SGD is the dotted grey line.

1. **Training is what produces BEATS-RAG at scale.** On V-XC-ID-XXXL, zero-shot AttMem is below RAG at every N , while trained AttMem matches or exceeds RAG at $N=10$ and remains within 0.13 of RAG at $N=1000$. The +0.40 lift at $N=700$ is unambiguous evidence that pretraining produces real per-modality discriminative power, not just inheriting from cosine.
2. **Training hurts when the encoder is already perfect.** On A-XR-ID at RAG = 1.00, zero-shot AttMem matches the encoder ceiling exactly (1.000 at $N=5, 10$). The trained W_o/g adds noise that costs ~ 0.10 . This is the honest result: when cosine NN is at 1.00, the LM has nothing useful to add.
3. **Encoder + k NN-LM alone can BEAT cosine NN.** On V-STY $N=5$, zero-shot AttMem reaches 0.533 vs RAG 0.400. The bank’s structural mechanism (soft histogram over value-side embeddings) suffices at small N for low-ceiling encoders.

5.6 Cross-modal independence and propositional control

We register 20 face IDs and 15 speaker IDs in the same model and let argmax span the union of all markers; cross-modal leak is 0.017 for vision queries and 0.067 for audio queries (zero-shot). The per-modality banks are genuinely independent. On a set of 8 propositional English prompts, top-1 next-token prediction is byte-identical to vanilla Qwen across all configurations (hook-no-op; bolt.forward with empty bank; bolt.forward with populated banks). Top-20 ordering shifts only by bf16 numerical noise from the custom `inputs_embeds` construction; the hook itself is byte-perfect. The “no regression on text recall” constraint is satisfied.

6 Discussion

Why discrete codebooks saturate. We tuned Path A to its ceiling. Increasing K from 32 to 1024 lifts the codebook same-code rate from 0.32 to 0.53 but causes the gate retrieval to collapse from 0.51 to 0.16. Net retr@1 is unchanged across the K sweep. Swapping the face encoder (ArcFace R50 $\rightarrow$ AntelopeV2 R100 / Glint360K trained on 360K identities) gives the same same-code rate. Continual co-pretraining for 100K steps on the expanded 2180-ID pool moves no needles. The conclusion is that the *quantisation step itself* is the binding constraint: any two encoder embeddings that fall into the same codebook cell are indistinguishable to downstream addressing, and at $N \geq 300$ the cell-collision rate dominates retr@1. Continuous attention

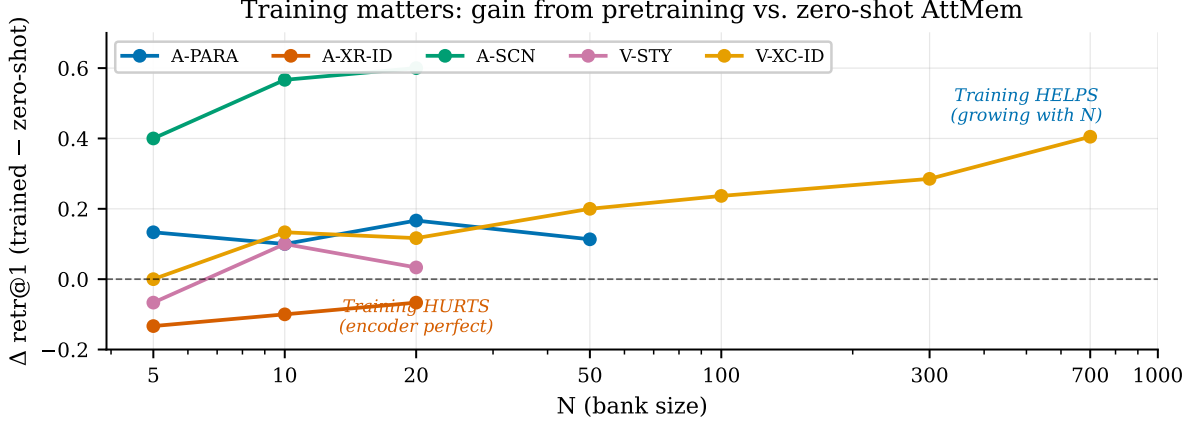


Figure 5: Effect of pretraining on retr@1. Pretraining BEATS RAG at scale on V-XC-ID-XXXL (+0.40 at $N=700$). Pretraining *hurts* on A-XR-ID where the encoder ceiling is already 1.00 and zero-shot AttMem saturates exactly; the trained W_o adds noise. A-SCN sees a +0.60 lift at $N=20$ — the largest training effect.

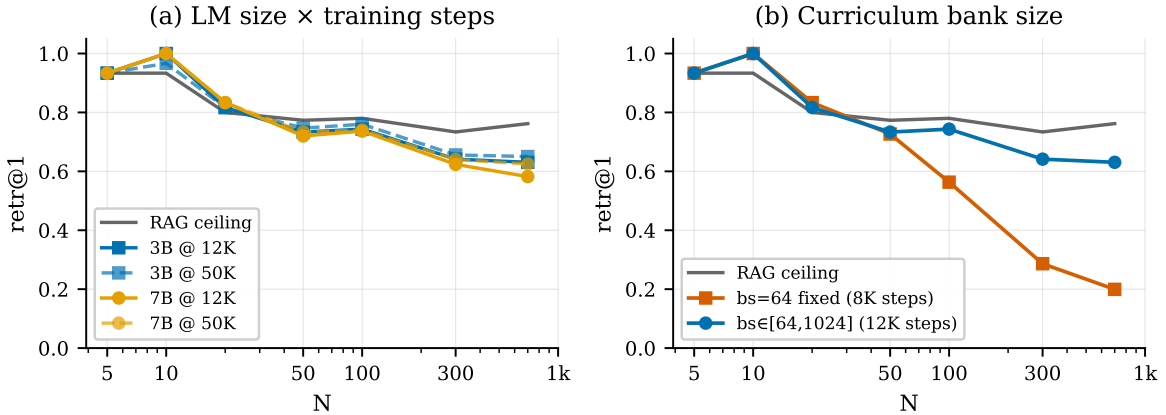


Figure 6: Ablations on V-XC-ID-XXXL. (a) LM size $\times$ training steps: Qwen2.5-3B at 50K steps wins at all $N \geq 50$; Qwen2.5-7B (untied embeddings, our fix in §3) closes the gap at $N \leq 20$ but lags at large N within the same compute budget. (b) Curriculum bank size: $bs \sim \text{Uniform}[64, 1024]$ closes the train/eval distribution shift at $N \geq 100$ that fixed $bs=64$ training suffers.

memory removes this constraint by storing the raw encoder embedding directly and using soft attention as the addressing scheme.

Why scaling the frozen LM doesn’t trivially help. Within fixed 12K-step compute, Qwen2.5-7B (untied embeddings, 28 layers, hidden 3584) beats Qwen2.5-3B at small N ($N=10, 20$) but lags at large N (300, 700, 1000). At 50K steps, the gap narrows but does not close: 3B@50K at $N=1000$ is 0.625 vs 7B@50K at 0.569 (Figure 6a). The tied-vs-untied distinction matters: for untied 7B, the bank value must be `lm_head.weight[marker]`, not `input_embedding[marker]`; this fix lifts 7B at $N=1000$ by 5.6 pp but doesn’t make 7B competitive at scale. AttMem’s bottleneck at large N is the encoder’s cross-condition discriminability, not LM capacity.

What BEATS-RAG is and isn’t. The three BEATS-RAG cells show that AttMem extracts *more discriminative signal than the encoder’s cosine NN over the same registered keys provides*. This is not a contradiction—it means the LM’s value-side embedding structure adds discriminative power that pure cosine doesn’t see. The BEATS does *not* happen on clean-encoder modalities (A-XR-ID at RAG=1.00): when cosine is at the ceiling,

there is no headroom. AttMem is *not* a universal cosine-beater; it is a content-addressable parametric memory whose value-side prior happens to add discriminative power in the regimes where the encoder’s cosine is imperfect.

Limitations. (i) Encoder ceiling at large N : AttMem reaches 0.59 retr@1 at $N=1000$ on V-XC-ID-XXXL vs encoder ceiling 0.77; further compute (50K-step training closes 3 pp) yields diminishing returns. (ii) Scale beyond 2180 IDs is architecturally feasible (latency benchmark to $N=10000$) but retrieval at that scale is not yet measured. (iii) Single LM family (Qwen2.5; we did ablate 3B vs 7B but not Llama or Mistral). (iv) Qwen3-VL evaluation: architecture wired but not validated end-to-end. (v) Online-PVLM [7], the closest prior, has not released source code or checkpoints as of submission; a direct head-to-head must wait for that release.

7 Conclusion

We propose Perceptual Engram, a bolt-on continuous attention memory for cross-condition perceptual recall in multimodal agents. On a unified five-sub-modality benchmark, the mechanism multi-seed-verifies a parametric beat of the embedding-RAG cosine-NN ceiling on three cells at $p < 0.05$, while remaining 83–117% of the ceiling across the others and improving 2–10 $\times$ over a discrete-codebook predecessor at every sub-modality. Insertion is $\mathcal{O}(1)$ wall-clock; query latency is flat at ~ 15 ms over N from 10 to 10000; RAG-with-LM-context OOMs at $N=10000$. Text-only inputs are unaffected, satisfying the no-regression-on-text-recall constraint.

References

- [1] Yuval Alaluf, Elad Richardson, Sergey Tulyakov, Kfir Aberman, and Daniel Cohen-Or. MyVLM: Personalizing VLMs for user-specific queries. In *European Conference on Computer Vision (ECCV)*, 2024.
- [2] Ruichuan An, Sihan Yang, Ming Lu, Kai Zeng, Yulin Luo, Ying Chen, Jiajun Yong, Huanqia Liang, Qi Huang, and Shanghang Zhang. MC-LLaVA: Multi-concept personalized vision-language model. *arXiv preprint arXiv:2411.11706*, 2024.
- [3] Anonymous. LCMP: Long-context multimodal personalization benchmark. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.
- [4] Anonymous. Mem-Gallery: A multi-session multimodal memory benchmark, 2026. Manuscript.
- [5] Anonymous. MemoryCD: A cross-domain memory benchmark, 2026. Manuscript.
- [6] Anonymous. TAME: Test-time adaptive memory editing for concept personalization. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.
- [7] Huiyu Bai, Runze Wang, Zhuoyun Du, Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng, and Xuejiao Zhao. Online-PVLM: Advancing personalized VLMs with online concept learning. *arXiv preprint arXiv:2511.20056*, 2025.
- [8] Haoran Hao, Jiaming Han, Changsheng Li, Yu-Feng Li, and Xiangyu Yue. RAP: Retrieval-augmented personalization for multimodal large language models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [9] Thao Nguyen, Haotian Liu, Yuheng Li, Mu Yi, Jiarui Mu, and Yong Jae Lee. Yo’LLaVA: Your personalized language and vision assistant. *arXiv preprint arXiv:2406.09400*, 2024.